



Natural Language Systems

Tutorial I: Project

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Topic Today

- Project

Project

- You will build a voice assistant that is able to provide weather information and manage calendar entries
- You will work in a team of 4-5 students
- All processing has to happen locally, i.e. you cannot use cloud models
- The final (graded) delivery will be a docker container that contains your voice assistant + an evaluation report
- There will be additional Milestones that help you to keep track of your progress, however they will not be graded

Milestones

- MS 1: Working code for ASR and TTS (14.11.2025)
- MS 2: Working code to connect to the weather and calendar API (28.11.2025)
- MS 3: Working voice assistant (12.12.2025)
- MS 4: Final Docker container and evaluation report (30.01.2025)

Requirements

- The system is able to take spoken English as input and produces spoken English as output
- The system keeps track of the conversation history
- The system does not connect to any other external resource than the weather and calendar API
- The user can request all information provided by the weather API through spoken language
- The user can perform all CRUD operations provided by the calendar API
- The user can refer to the previous turn in the conversation

Weather API

Sample request:

```
curl -X POST -d "place=Marburg" https://api.responsible-nlp.net/weather.php
```

Sample response:

```
{
  "place": "Marburg",
  "forecast": [
    {
      "day": "thursday",
      "temperature": { "min": 7, "max": 15 },
      "weather": "few clouds"
    },
    {
      "day": "friday",
      "temperature": { "min": 6, "max": 14 },
      "weather": "rain"
    },
    {
      "day": "saturday",
      "temperature": { "min": 5, "max": 11 },
      "weather": "clear sky"
    }
  ]
}
```

Weather API

- The weather API always returns a forecast for the next 7 days (including the current day)
- The following weather conditions are available:
 - "clear sky"
 - "few clouds"
 - "scattered clouds"
 - "broken clouds"
 - "shower rain"
 - "rain"
 - "thunderstorm"
 - "snow"
 - "mist"

Calendar API

CREATE (POST)

```
curl -X POST "https://api.responsible-nlp.net/calendar.php" \  
  -H "Content-Type: application/json" \  
  -d '{"title":"Meeting with team","description":"Discuss  
project","start_time":"2025-11-03T09:00","end_time":"2025-11-  
03T10:00","location":"Room 12"}'
```


Calendar API

List all (GET)

```
curl "https://api.responsible-nlp.net/calendar.php"
```

Calendar API

Get single (GET):

```
curl "https://api.responsible-nlp.net/calendar.php?id=1"
```

Calendar API

Update (PUT):

```
curl -X PUT "https://api.responsible-nlp.net/calendar.php?id=1" \  
  -H "Content-Type: application/json" \  
  -d '{"title": "Updated title", "location": "Room 15"}'
```

Calendar API

Delete (DELETE):

```
curl -X DELETE "https://api.responsible-nlp.net/calendar.php?id=1"
```

Required Commands

- The system must be able to (among others) correctly process and answer the following commands:
 - What will the weather be like today in Marburg?
 - What will the weather be on Friday in Frankfurt?
 - Will it rain *there* on Saturday?
- Where is my next appointment?
- Add an appointment titled XYZ for the 12th of January.
- Delete the previously created appointment.
- Change the place for my appointment tomorrow.

Evaluation Report

- The evaluation report should be at most 2 DIN A 4 pages + title sheet containing all team members (name + matriculation ID)
- You decide how you evaluate your system based on the content of the lecture